



UC San Diego

Electrical and Computer Engineering
JACOBS SCHOOL OF ENGINEERING

DIGI
HEALTH
LAB

Machine Learning for Swallow Analysis

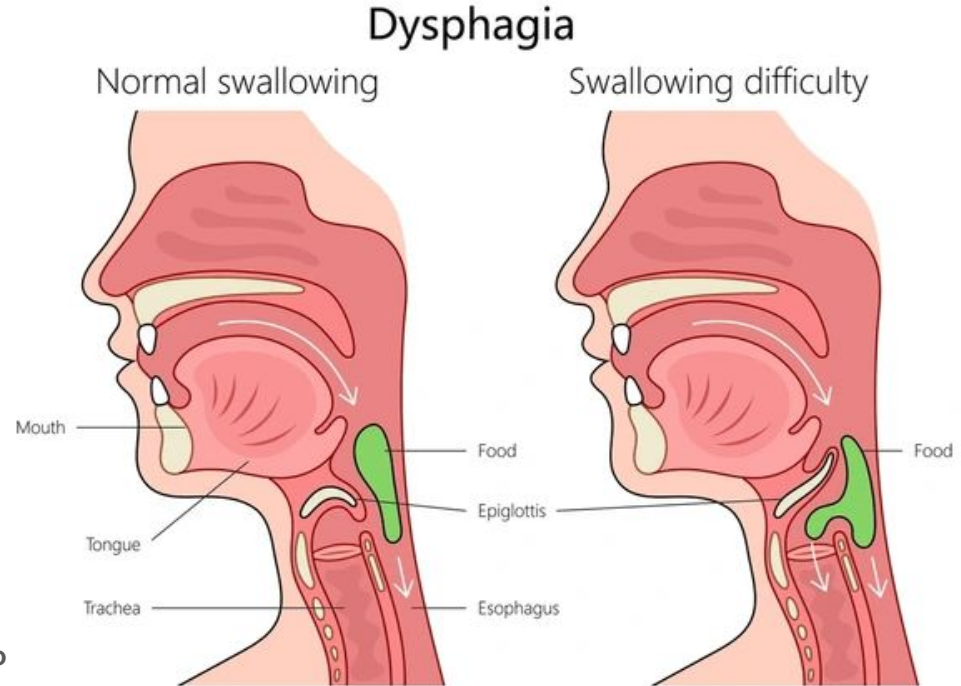
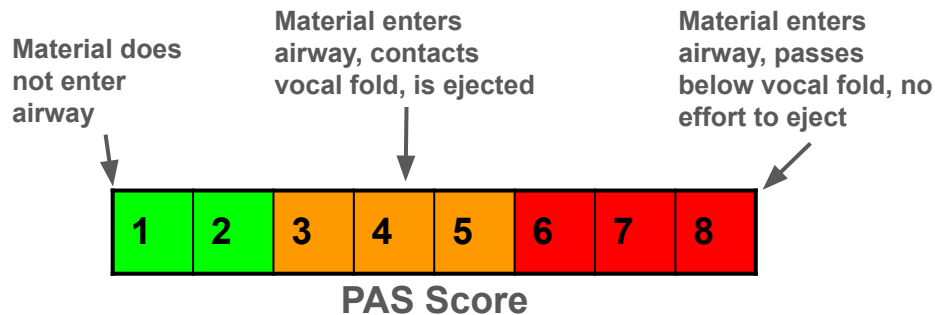
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UCSD SRIP

Agenda

1. Background/Objective
2. Project Framework
 - a. CNN Segmenter
3. Results
4. Conclusion
5. Acknowledgements
6. Q&A

Background

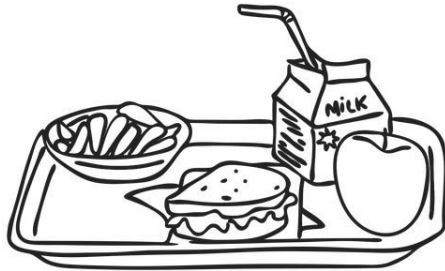
- **Some Patients swallow abnormally**
 - This can be dangerous
- **Current screening methods**
 - Expensive (endoscopy)
 - Unreliable (evaluation by sight)



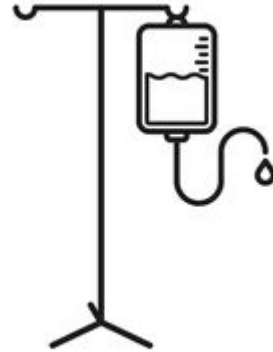
Objective

- Develop an audio processing system to classify swallowing sounds as dangerous or safe for patient screening.

Use Case:

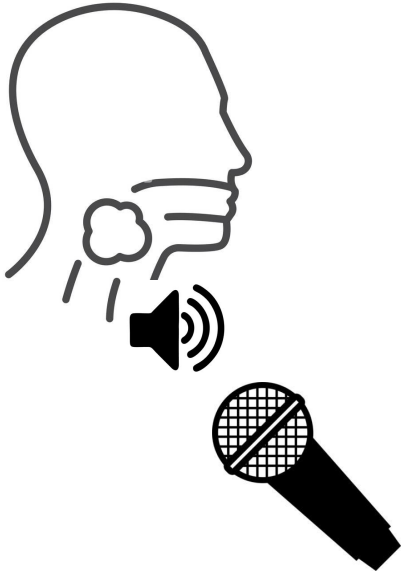


or

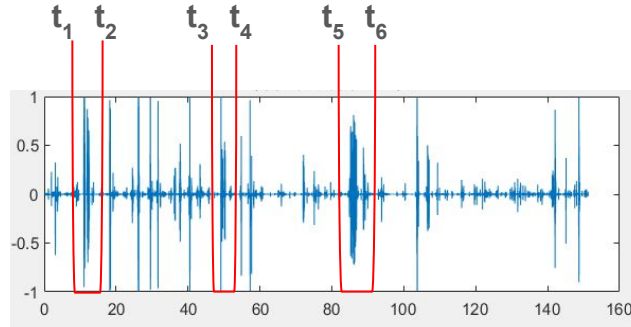


Framework

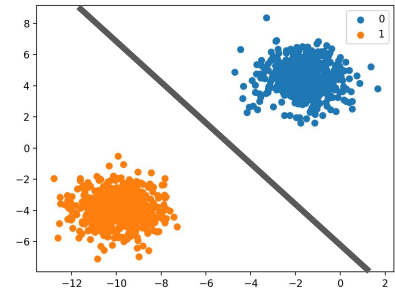
1. Record Swallowing Sound



2. Segment swallows



3. Extract Features and Classify Swallows

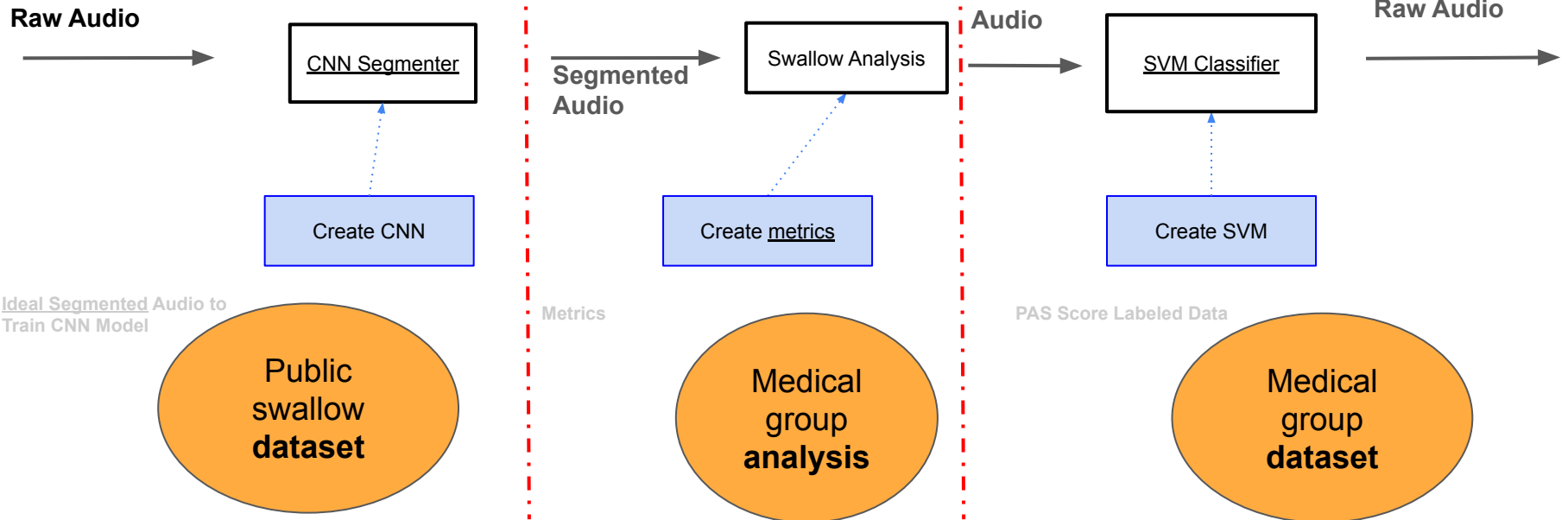
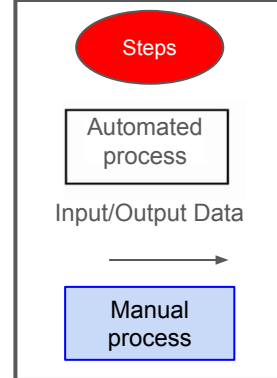


Framework: Audio to Classification

1

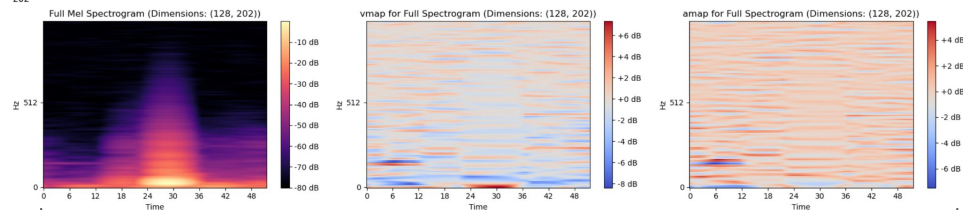
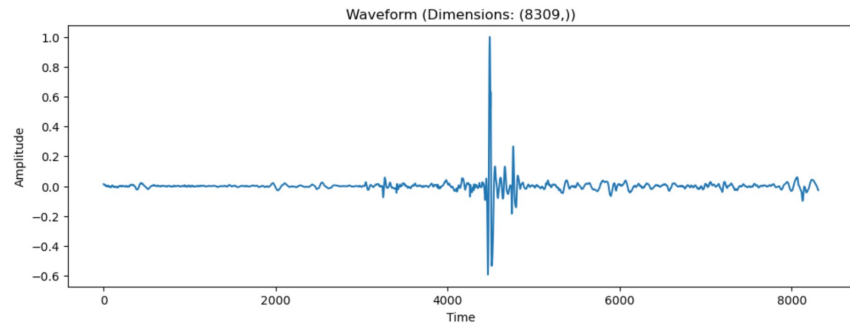
2

3



CNN Segmenter

- Timestamps swallows in input audio file.
- Training with SeMG Dataset: 110 timestamped audio files



Labeled 3x128x128 images from 2 second segments

Resnet
(add image)

} 0: other
1: swallow

Feature Extraction

- Gather features (from medical group): Number of Swallows, Swallow Length(s), Peak Amplitude, Average Amplitude, Amplitude^2, Area under curve...

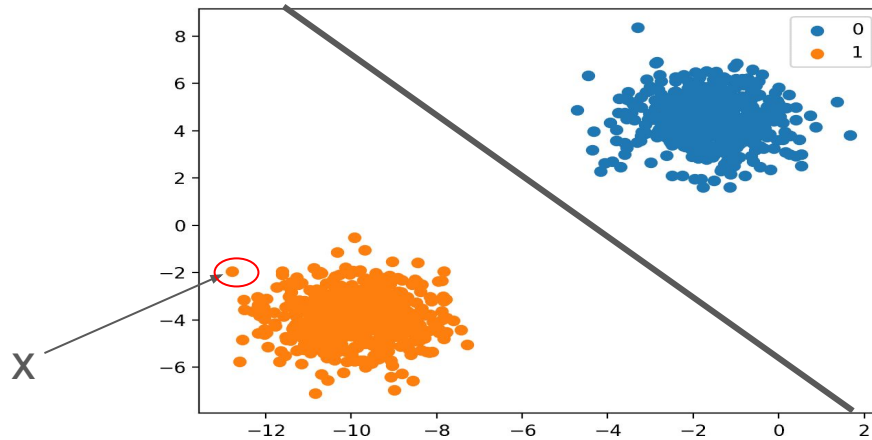
feature vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix}$

features 1 to n

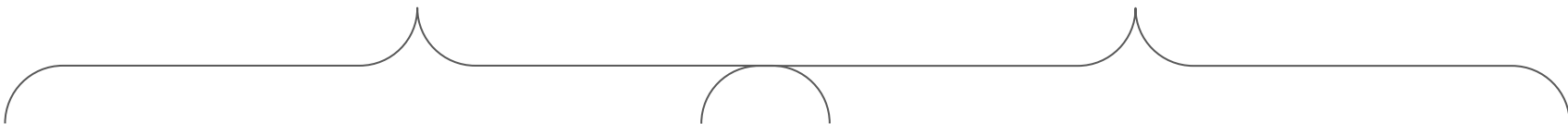
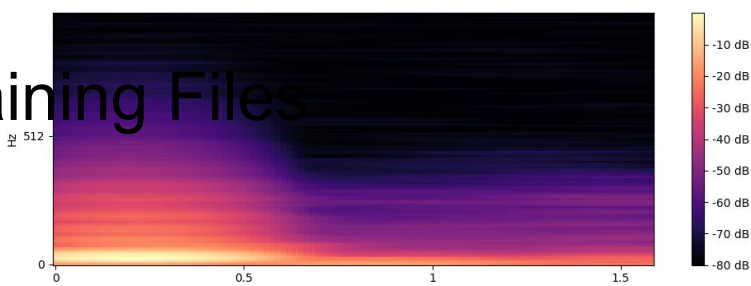
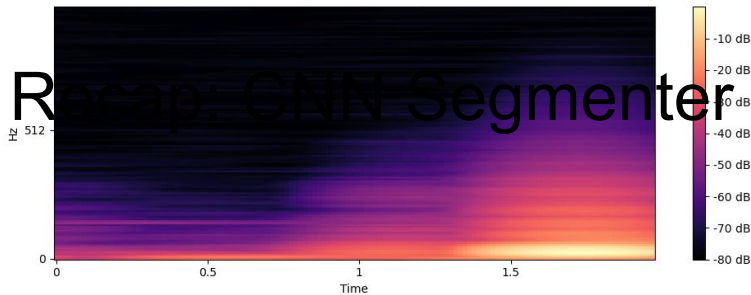


SVM

- Create an SVM classifier from features
 - Each audio file given by the medical group are **labeled** from a scale if 1-8



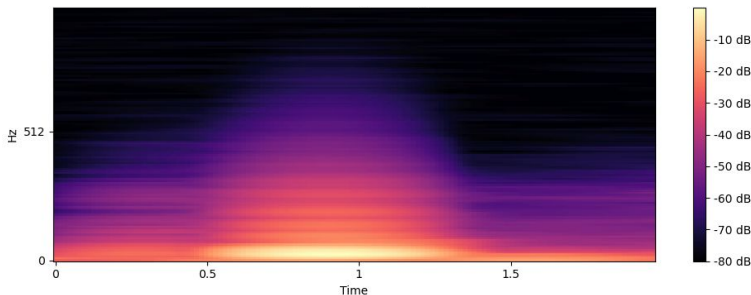
Recap: CNN Segmenter Training Files



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4 seconds of audio (100ms timestamps)

Label 1



Issue with accurate spectrogram timestamping relative to the original label vector

Hyperparameters:

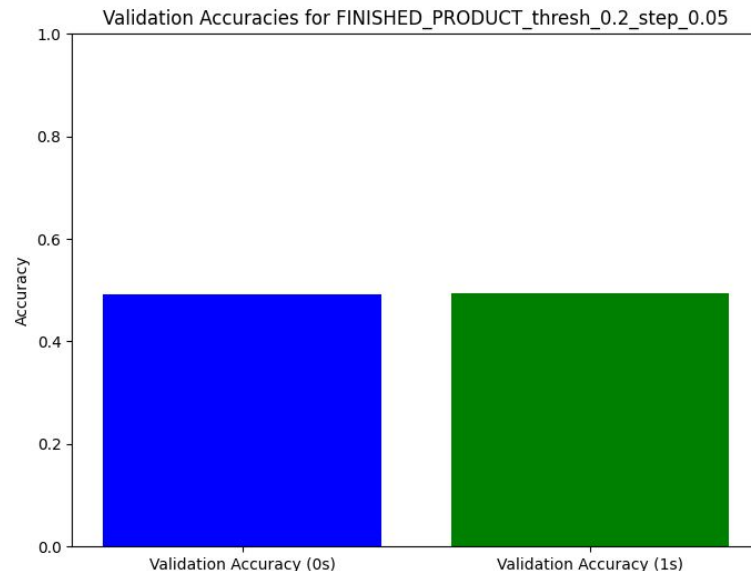
1. min # of 1s in segment to label as 1 ($\geq 30\%$)
2. Jump distance (100ms)

Results: CNN Classifier (step 1)

- Challenges in consistent class differentiation.

Next Steps:

- Modify feature selection
 - Focus Spectrograms on Swallows
 - Use Wavelet-Transform to analyse spectrum and create better spectrograms
- Modify model training
 - Implement regression helper task for model improvement

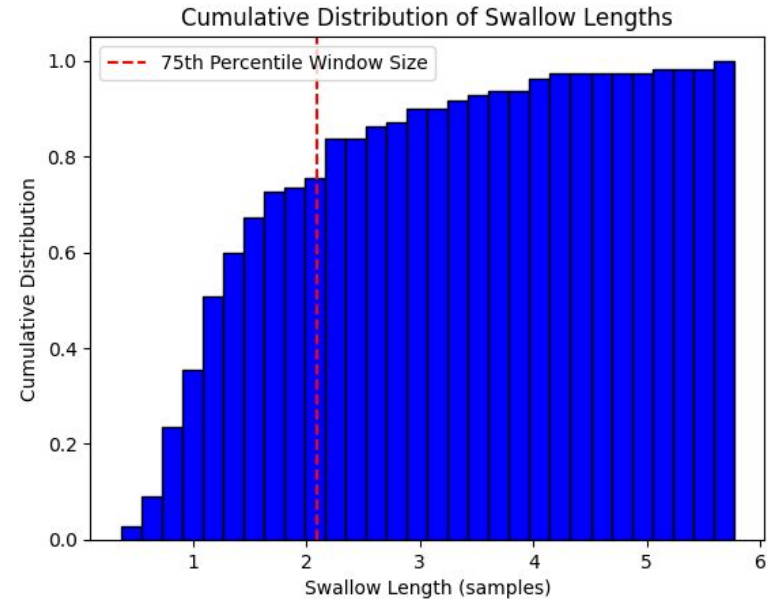
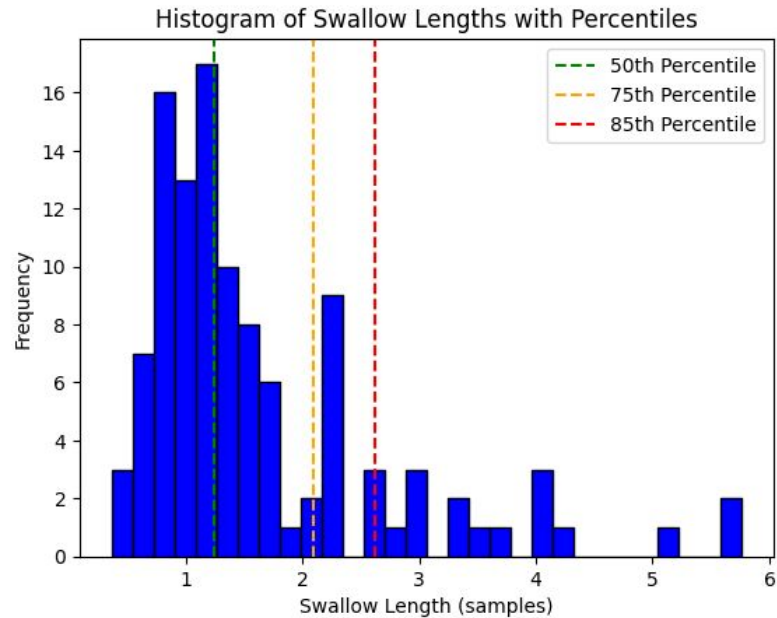


Conclusion

- Classification task
 - Refining data and model to improve accuracy
- Significance of research
 - Cost-effective, non-invasive tool, Broad implications for medical diagnostics
- Future Work
 - Implementing regression helper task
 - Enhancing feature extraction and model performance

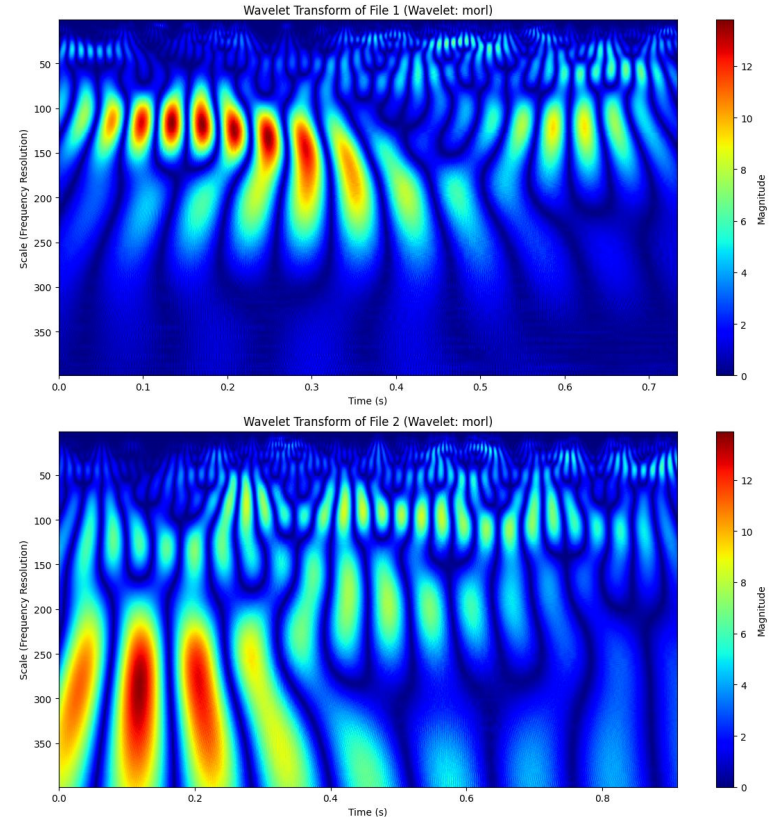
Average Swallow Length in SeMG dataset

- Recommended window size 2 sec.



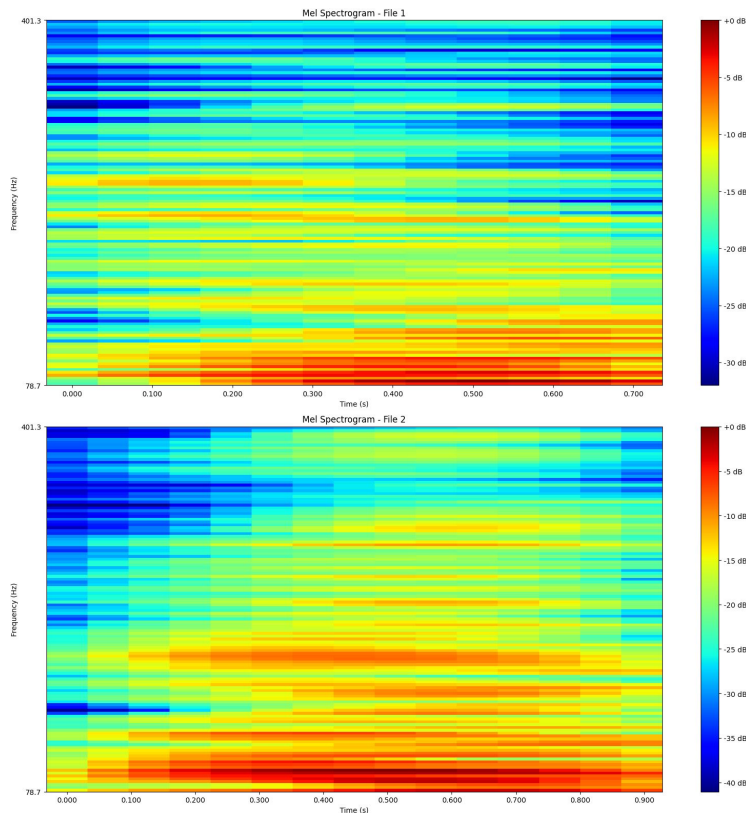
Wavelet Transform using same scale

- **Wavelet Transform:** Applied continuous wavelet transform (CWT) to compare multiple swallow audio samples, revealing variance in energy concentration across different frequency bands. Identified two key scale ranges that showed changes: 80 - 200 and 200 - 350 scales
- Future work: overlay labels with Wavelet transform
 - Check the Center frequency
 - ft of all swallows and calculate centroid or mean
 - Create distribution of all swallows for centering - use that to set
 - Quick experiment - compare results



Mel Spectrograms Comparison

- **Spectrogram Analysis:** In the mel spectrograms, we expect there to be frequency changes in [insert here].



Analysis Conclusion

- Optimal frequency range for creating mel-spectrograms is [insert here]

Acknowledgements

- Prof. Edward J. Wang
 - PI/Mentor
- Nancy Liu
 - Mentor
- Yunfei Luo
 - Mentor

Q&A